EE 2001 Experiment 2

Group no: 2.3

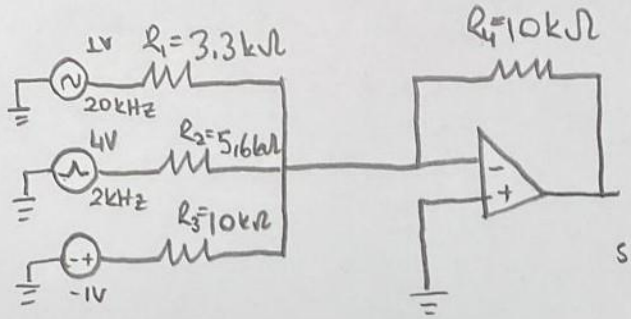
Group member 1: Abdussamet Şahin , 150718017 ,
Construct the circuit, adjusting oscilloscope and function generators

Group member 2: Mehmet Ali Kanalak, 150720062 ,
calculating, construct the circuit, preparing lab report,

Group member 3: Hüseyin Aydın Tokhan, 150715661 ,
calculating output voltage and determining the resistor, construct the circuit.

1

a)



As we know that the formula of opam's gain : $\frac{R_4}{R_{in}}$

so for 3 gain : $3 = \frac{R_4}{R_1}$

for 2 gain : $2 = \frac{R_4}{R_2}$

for 1 gain : $1 = \frac{R_4}{R_3}$

We can select the resistor values only on 10k, 5,6k, 3.3k, 2,2k. So we can choose R_4 as a 10k and the other resistor values will be =>

R_1 : 3,3k

R_2 : 5,6k

R_3 : 10k

because the closest result we will get from these values.

b) For determining opamp feeding voltages, we should consider the formula of : $-V_{cc} < V_o \leq +V_{cc}$

Teoritically we can get V_o as :

$$\frac{1}{3.3k} + \frac{4}{56k} + \frac{-1}{10k} = \frac{-V_o}{10k} \Rightarrow V_o = -9,17V$$

So we should feed this opamp +15 V_{cc} -15 V_{cc} is ok. It is also suitable with the opamp's datasheet.

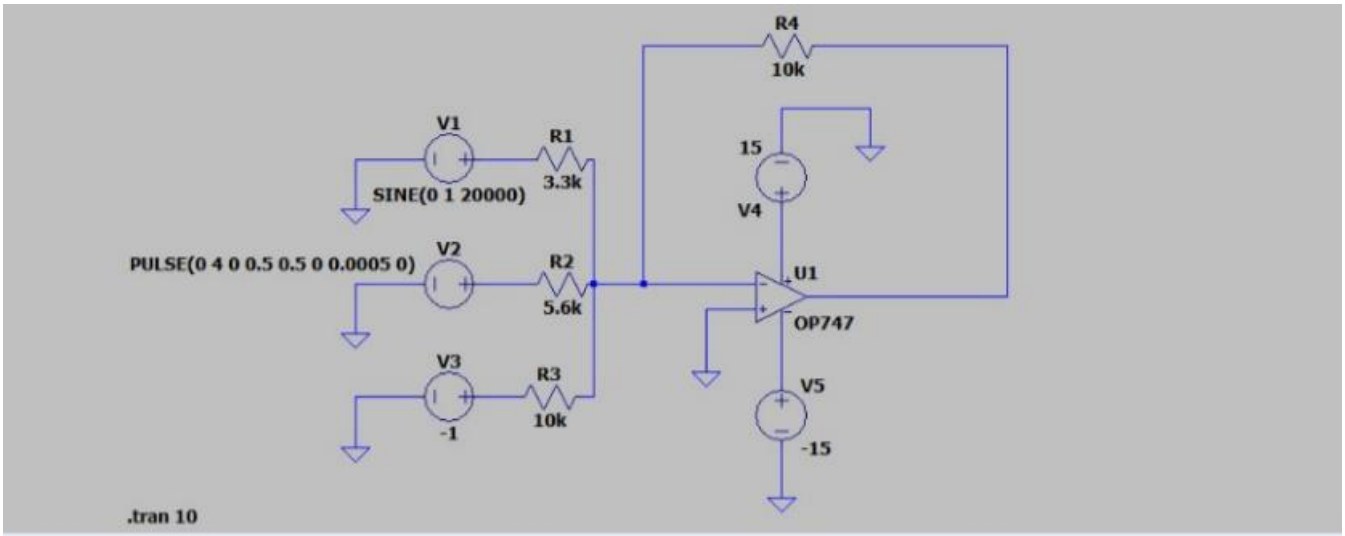


Figure 1: 1a) Determining resistor values

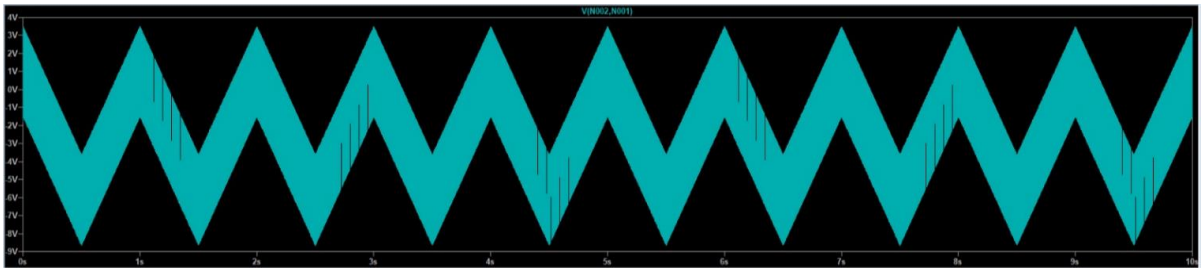


Figure 2: 1c) Guess and sketch output signal.

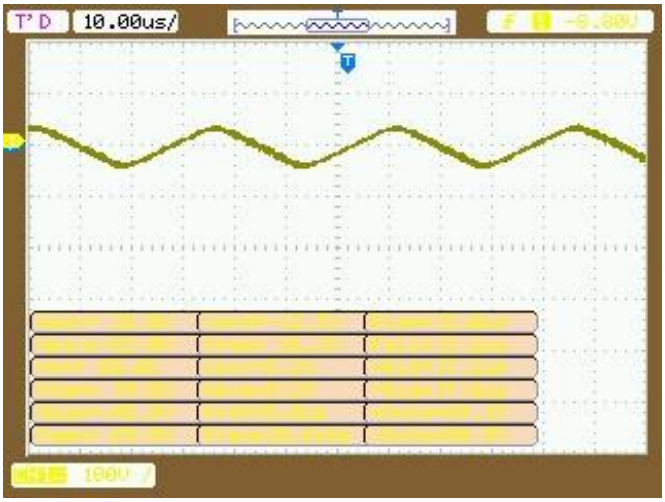


Figure 3: 1d) Construct the circuit, record input and output voltages.

Sinusoidal Signal (input 1): The frequency at the output will be the same (2kHz), and the amplitude will be three times the input amplitude (3V).

Triangular Wave (Input 2): The frequency at the output will be the same (2kHz), and the amplitude (12V).

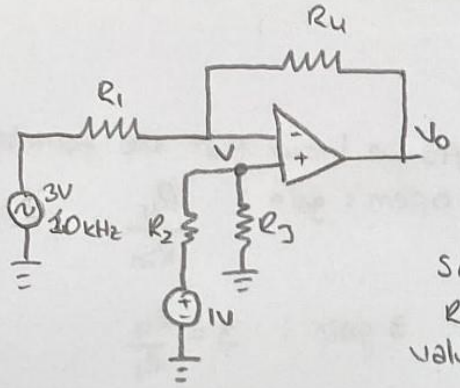
DC signal (input 3): The DC signal will contribute negatively, providing and offset to the output.

The output waveform will be a combination of these three signals, with the amplitude of the sinusoidal and triangular waves being positive and the amplitude of the DC signal being negative. The frequencies of the sinusoidal and triangular waves will remain the same at the output.

Figure 4: 1e) Show frequency and amplitude contributions of each input component at the output waveform.

2

a)



c) As we know the gain of a opamp is: $\frac{R_4}{R_1} = \frac{V_o}{V_i}$ so $2V_i = \frac{V_o}{V_i}$

$\Rightarrow V_o = 2V_i$

So we can choose $R_4 = 10k\Omega$ and $R_1 = 5{,}6\ k\Omega$ from the resistor values.

b) $-V_{cc} \leq V_o \leq +V_{cc}$

Theoritically $V_o \Rightarrow$ $\frac{3V - V}{R_1} = \frac{V - V_o}{R_4}$ and we choose $R_1 = 5{,}6k$, $R_2 = 3{,}3k$, $R_3 = 2{,}2k$, $R_4 = 10k$

$V_o = -5{,}04V$ so $V = 0{,}4V$

So from the datasheet and the equation we can feed the opamp with $V_{cc} = +15$, $V_{-cc} = -15$ is ok.

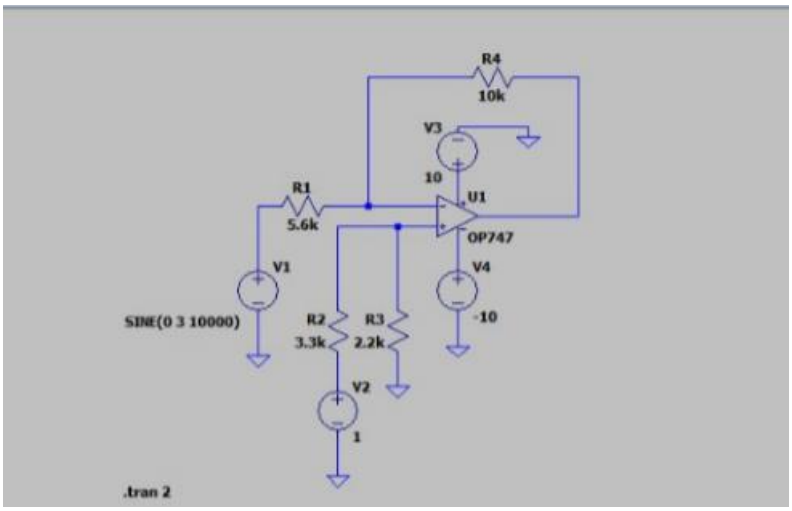


Figure 5: 2a) Determining resistor values

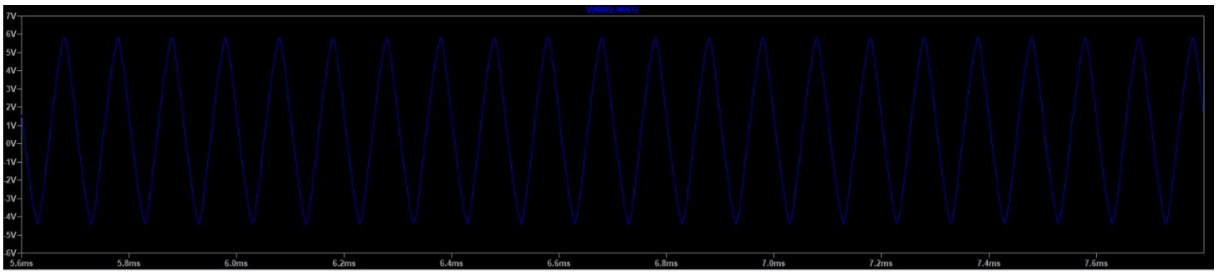


Figure 6: 2c-d) Guess and sketch output signal

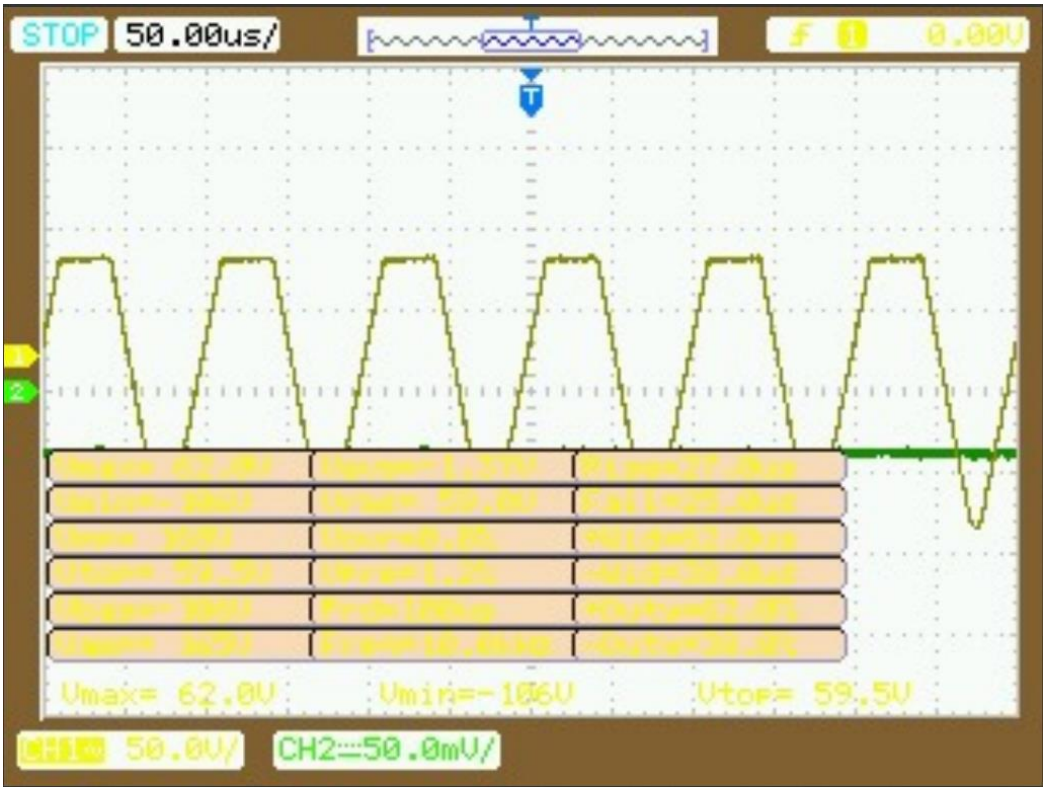


Figure 7: 2-e) Construct the circuit, record input and output voltages.

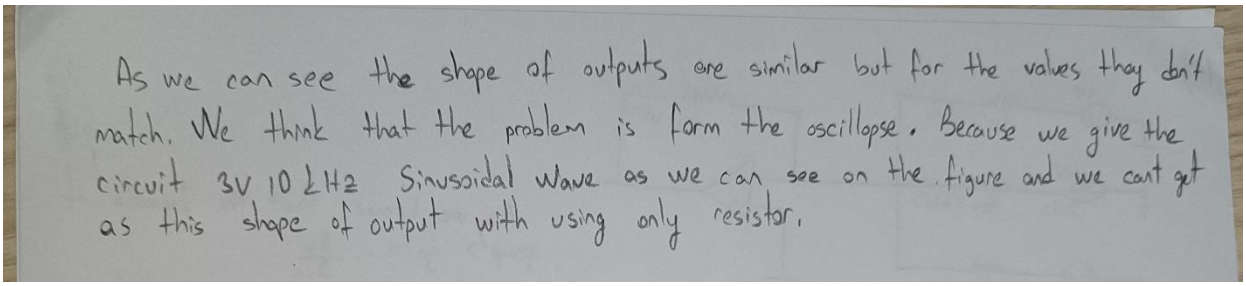


Figure 8: 2-e



Figure 9: 2-e

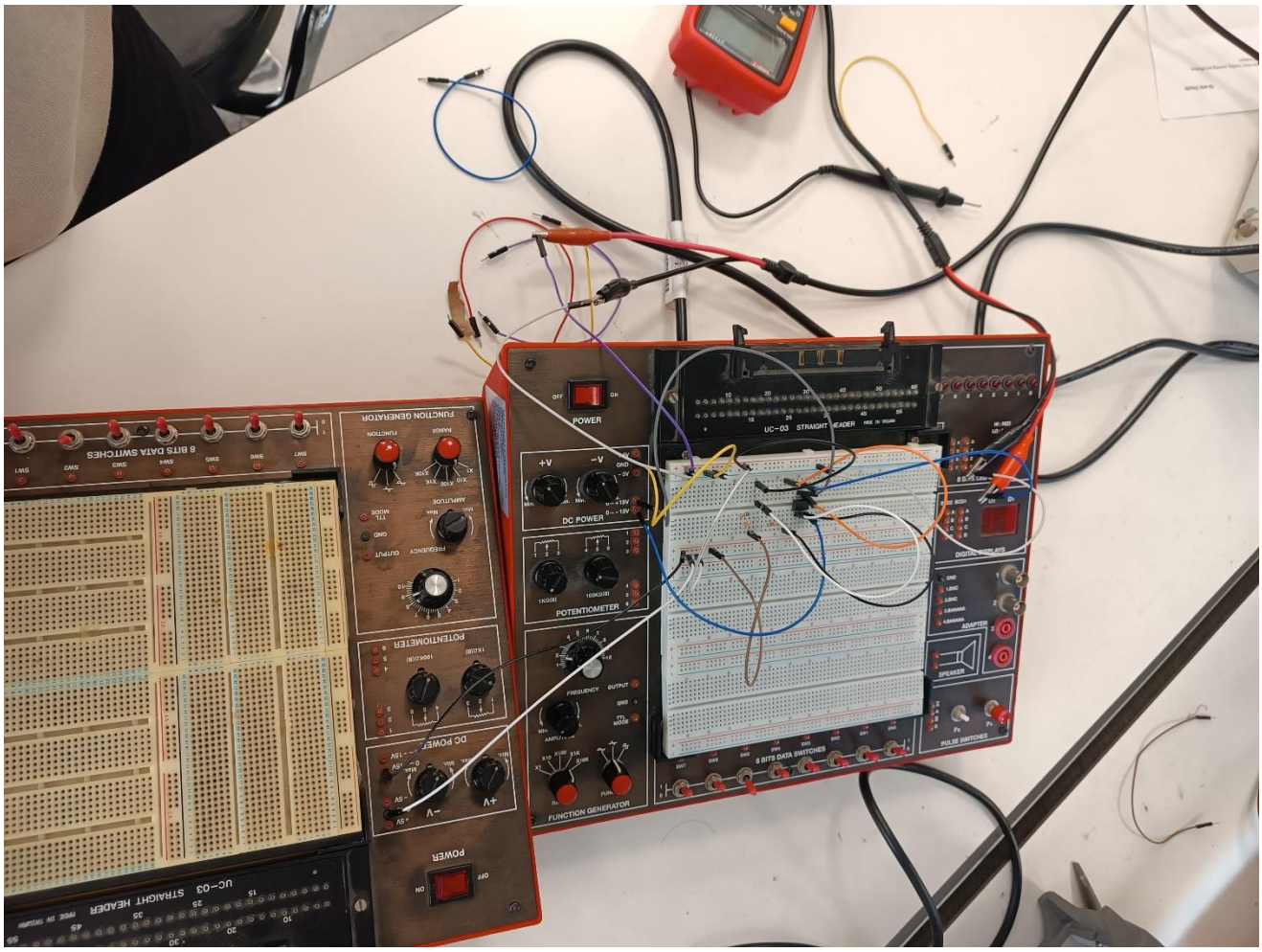

POWER
DC POWER
POTENTIOMETER
FUNCTION GENERATOR
8 BITS DATA SWITCHES
UC-03 STRAIGHT HEADER
SPEAKER
ADAPTER
PULSE SWITCHES

Figure 10: 2-e